Artificial Intelligence for Predictive Maintenance: A Machine Learning Approach using Publicly Available Manufacturing Data for Kenya's Industry

1. Project Context, Goals, and Requirements

This report details the comprehensive technical methodology required to implement a robust, data-driven Predictive Maintenance (PdM) solution. The project utilizes the publicly available AI4I 2020 dataset [1], a synthetic, multivariate time-series dataset that accurately reflects condition monitoring data encountered in the manufacturing sector.[2] The objective extends beyond standard model development to ensure the solution is directly relevant and strategically advantageous for industrial operations, specifically within Kenya's manufacturing and agro-processing sectors.[3]

The adoption of PdM represents a strategic shift from reactive or scheduled preventive maintenance toward proactive, Condition-Based Monitoring (CBM) and Prognostics and Health Management (PHM).[4] For Kenyan industry, this transformation is crucial, as the primary operational goals are to minimize massive production losses from unplanned downtime and reduce maintenance costs associated with the high reliance on imported machinery, raw materials, and spare parts.[3]

1.1 Dual Predictive Maintenance Objectives

To achieve complete operational intelligence, the project requires the development of two interconnected machine learning pipelines:

1. **Failure Classification Pipeline:** This model provides a binary or multi-class classification, predicting the imminent occurrence of a failure event (e.g., within the next operational window). This supports urgent, short-term maintenance intervention.[5]
2. **RUL Prognosis Pipeline:** This model provides a continuous regression estimate of the Remaining Useful Life (RUL), quantifying the expected time (in cycles or hours) before a machine requires repair or replacement.[7] This is essential for long-term optimization of scheduling and inventory management.[4]

1.2 Foundational Data Characteristics

The AI4I 2020 dataset consists of 10,000 data points with 14 features.[1] Key condition monitoring features include continuous variables like Air Temperature [K] and Process Temperature [K], and integer/continuous variables such as Rotational Speed [rpm], Torque [Nm], and Tool Wear [min].[1] The dataset also includes a categorical feature representing product quality variants (L, M, or H).[1]

Critically, the dataset targets machine failure, which includes five specific failure modes: Tool Wear Failure (TWF), Heat Dissipation Failure (HDF), Power Failure (PF), Overstrain Failure (OSF), and Random Failures (RNF).[1] Initial analysis confirms the dataset is highly imbalanced [9], with failure events representing a minority class, such as only 98 datapoints exhibiting Overstrain Failure.[8] This imbalance dictates the technical approach to modeling and evaluation.[10]

2. Data Engineering and Contextualization

The first technical phase focuses on establishing a foundation of clean, contextually relevant data suitable for both classification and prognostics.

2.1 Data Sourcing, Validation, and Causal Discovery

Data acquisition begins by securing the 10,000 instances [1] and verifying the feature types and distributions. A crucial step beyond basic data cleansing is the deep structural analysis of the failure types.

The documentation for the AI4I 2020 dataset provides explicit causal definitions for certain failures. For instance, the Overstrain Failure (OSF) is not merely correlated with high sensor values, but is explicitly defined to occur if the product of Tool wear and Torque exceeds a specific, variant-dependent threshold (e.g., 11,000 minNm for L products).[8]

Standard machine learning models typically focus on statistical correlation. For a highly robust and trustworthy predictive model, the project must move beyond standard feature selection and incorporate known physics into the data preparation. By explicitly calculating a **Stress Index** (e.g., Torque $\times$ Tool Wear), the resulting feature directly captures the known causal mechanism driving mechanical degradation.[1] This engineering-grounded approach fundamentally strengthens the explainability of the final model and ensures the predicted failure is traceable to a verifiable physical condition.

2.2 Contextual Mapping to Kenyan Industrial Assets

The synthetic dataset models a milling machine.[8] To ensure the PdM solution is actionable for Kenyan industries, the sensor features must be logically mapped to the operational characteristics and failure modes of common agro-processing and manufacturing machinery (e.g., centrifugal pumps, crushers, industrial mixers, and dryers).
Table 1 illustrates the critical mapping of generic sensor features to the condition indicators of relevant machinery found in the target sectors.
Table 1: Mapping AI4I 2020 Features to Kenyan Industrial Equipment and Failure Modes

AI4I 2020 Feature	**Likely Sensor/Reading**	**Relevant Kenyan Machinery Type (Agro/Manufacturing)**	**Common Failure Mode Predicted**
Air/Process Temperature [K]	Thermocouple, RTD	Industrial ovens, Grain dryers, Cooling systems, Extruders	Heat Dissipation Failure (HDF), Material degradation, Sensor soft faults [12]
Rotational Speed [rpm]	Tachometer, Encoder	Centrifugal pumps, Gearboxes, Crushers, Mixers	Excessive Vibration, Bearing Seizure, Transmission damage
Torque [Nm]	Strain Gauge/Torque Sensor	Heavy-duty conveyers, Agitators, Hydraulic presses	Overstrain Failure (OSF), Mechanical binding, Power system fault
Tool Wear [min]	Time-based Counter	Milling tools, Dies, Molds, Shear blades (parts requiring replacement)	Premature Component Failure (TWF), Quality Deviation

A deep understanding of the operating environment is paramount for reliable PdM implementation. Agricultural and manufacturing environments in Kenya often subject equipment to long run times and challenging external conditions, including extreme heat and varied weather.[13] This harsh operating environment significantly increases the risk of not only machine failure but also **sensor soft faults**, such as signal drift, bias, or spike faults.[12] If the machine learning model is trained on, or deployed against, data contaminated by these sensor issues, the predictions will be fundamentally unreliable.[15]
Therefore, an essential pre-processing module must be integrated into the pipeline to perform **Anomaly Detection on the Sensor Data itself** prior to feeding the data into the predictive model. Techniques such as Isolation Forests or statistical process control should be employed to identify and flag potential sensor anomalies.[16] This guarantees that the primary PdM model diagnoses machine health issues and not faulty input streams, which is a critical necessity for

maintaining system integrity in demanding operating conditions.[15]

2.3 Comprehensive Feature Engineering

The multivariate time-series structure of the dataset [1] allows for the creation of features that capture temporal dynamics and degradation trends.[17]

2.3.1 Standardization and Derived Features

Continuous features (Temperatures, Torque, Speed) must be scaled using standardization or normalization to ensure uniform contribution during training. Categorical features like Type must be processed using One-Hot Encoding.[1]
Beyond the raw readings, the advanced features include:

1. **Degradation Trend Features:** Calculating rolling statistical metrics such as the mean, standard deviation, and the slope of the trend for key sensor variables (Temperature, Torque, Speed) over operation windows (e.g., 3-hour or 24-hour periods).[17] The positive slope of temperature, for example, is a strong condition indicator of impending HDF.
2. **Lagged Features:** Incorporating measurements from previous operational cycles to capture temporal dependencies (e.g., comparing the current torque reading to the torque reading 5 cycles prior).[18]
3. **Derived Physics Features:** Explicitly calculating the known causal features, such as the **Stress Index** (Torque $\times$ Tool Wear) and Power Input (Torque $\times$ Rotational Speed), provides the model with direct metrics on mechanical load.

2.3.2 RUL Target Generation and Dimensionality Management

For the Prognosis Pipeline (Section 5), the classification target must be converted into a continuous variable representing the **Remaining Useful Life (RUL)**.[7] Assuming the sequential nature of the UDI entries reflects time steps, the RUL at any time step t is calculated as $RUL_t = T_{failure} - t$, where $T_{failure}$ is the time step of the known or predicted failure event for that product run.[7] This transformation enables regression modeling.
The aggressive nature of feature engineering (rolling statistics, lags, derived variables) significantly expands the feature space, leading to a high-dimensional input vector. While this richness can be beneficial, it risks encountering the "curse of dimensionality," where sparsity affects the reliability of predictions.[20] To maintain model robustness and optimize computational efficiency—a necessary consideration for resource-conscious industrial environments—a step involving **Dimensionality Reduction** must be implemented. Techniques such as Principal Component Analysis (PCA) or feature selection based on correlation metrics should be used to project the data onto a lower, more reliable dimension, particularly for the

Prognosis Pipeline.[20]

3. Pipeline I: Failure Classification for Imminent Maintenance Decisions

The classification pipeline is designed to serve as the critical alarm system, detecting the high probability of imminent failure to prevent catastrophic unplanned downtime.

3.1 Model Selection and Imbalance Management

The chosen algorithms for classification are **eXtreme Gradient Boosting (XGBoost)** and **LightGBM**.[11] These ensemble methods, based on the gradient boosting framework, sequentially assemble multiple decision trees and are known for their high performance, speed, and computational efficiency.[21]

Given the severe class imbalance inherent in the AI4I 2020 dataset [9], the training strategy must explicitly address the rarity of positive failure samples. Two methods are available:

1. **Data Augmentation:** Applying techniques like Synthetic Minority Oversampling Technique (SMOTE) or exploring advanced Generative AI methods (e.g., CTAGN) to create synthetic instances of the minority failure class, thereby balancing the training data.[9]
2. **Weight Balancing:** Utilizing model-native mechanisms, such as setting the scale_pos_weight parameter in XGBoost, to assign higher weights to the positive (failure) instances during training. This is a common method when focusing on overall performance metrics like AUC or F-score.[10]

3.2 Hyperparameter Optimization and Metric Prioritization

Hyperparameter optimization (HPO) is essential to manage the bias-variance trade-off, controlling model complexity through parameters like max_depth, min_child_weight, and gamma to prevent overfitting.[10]

The most critical aspect of this pipeline is the selection of the evaluation metric, which must reflect the asymmetrical cost structure of industrial failure. The operational cost of a **False Negative** (a failure event predicted as normal operation, resulting in unplanned downtime) is exponentially higher than the cost of a **False Positive** (a normal event predicted as failure, resulting in unnecessary inspection).[4]

Therefore, the evaluation must prioritize the model's ability to detect true positive failures (Recall). The standard F1-score gives equal weight to Precision and Recall.[24] For this

application, the HPO must optimize for the **F-beta score ($\beta > 1$)**, where a beta value greater than 1 explicitly weights Recall more heavily than Precision.[24] This metric ensures that the model is primarily optimized to minimize unplanned disruption, maximizing reliability, while Precision serves as a secondary metric to control operational overhead and avoid the "cry-wolf" syndrome that erodes user trust.

Table 2: Selection Rationale for Predictive Maintenance Evaluation Metrics

Metric	Operational Implication in Kenyan Industry	Prioritization Rationale
Recall (Sensitivity)	Ability to detect critical failures, minimizing catastrophic unplanned downtime. Crucial given the high costs and time associated with acquiring imported spares.[6]	**Highest Priority.** Directly measures success in proactive maintenance strategy.
F-beta Score ($\beta > 1$)	Provides a robust, single-score assessment under imbalanced conditions. Explicitly values failure detection (Recall) more than false alarms (Precision).	**Primary Optimization Metric.** Reflects the high operational cost of failure in resource-constrained environments.
Precision	Accuracy of positive predictions. Minimizes wasted labor and unnecessary maintenance, controlling operational overhead.	**Secondary Priority.** Used to monitor system trust and maintenance efficiency.

4. Pipeline II: Remaining Useful Life (RUL) Prognosis

The RUL Prognosis pipeline shifts the objective from binary alarm to continuous lifetime estimation, enabling optimal maintenance scheduling and resource planning.[4]

4.1 Sequence Preparation and Architecture Selection

The data must be structured into multivariate time sequences, formatted as [samples, sequence_length, features], with the continuous RUL label assigned to each sequence.[26] This sequence format is essential for leveraging deep learning architectures that excel at learning temporal features.[27]

For industrial prognostics, advanced sequential models often outperform traditional Recurrent

Neural Networks (RNNs) or standard Long Short-Term Memory (LSTM) networks.[27] The recommended architecture is the **Convolution-based Long Short-Term Memory (CLSTM) Network**.[28] This novel deep learning structure integrates convolutional operations into both the input-to-state and state-to-state transitions of the LSTM unit. This hybrid approach simultaneously captures both time-frequency features (localized patterns identified by the CNN component) and long-term temporal dependencies (managed by the LSTM component).[28] The CLSTM network has demonstrated superior accuracy and computation efficiency in RUL prediction on rotating machinery compared to deep CNNs and deep LSTMs.[28]

4.2 Training and Operational Synergy

The model will be trained using Mean Squared Error (MSE) or Root Mean Squared Error (RMSE) as the loss function, standard metrics for quantifying the magnitude of regression error.[20] The primary evaluation metric is RMSE, which measures the accuracy of the prediction in the native units of RUL (e.g., cycles or hours remaining).
The two modeling pipelines—Classification and RUL Prognosis—must operate in strategic synergy to maximize operational value. The classification model acts as the **high-priority trigger**, signaling with high confidence *if* an immediate intervention is required. Conversely, the RUL model acts as the **scheduling optimizer**, quantifying *when* the intervention should occur. An optimal decision-support system uses the classification trigger for safety and emergency alerts, while RUL predictions refine the maintenance schedule to prevent excessive downtime and ensure that corrective maintenance is performed only when truly needed, thereby avoiding unnecessary costs and maximizing equipment lifespan.[3]

5. Explainable AI (XAI) and Decision Support

Model interpretability is not merely an academic requirement; it is a prerequisite for achieving operational trust and enabling informed decision-making among maintenance managers, especially when introducing AI into established industrial practices.[29]

5.1 XAI Methodology Selection

Given the reliance on complex, high-performance models (XGBoost, CLSTM), the most effective methodology for achieving both local and global interpretability is **SHapley Additive exPlanations (SHAP)**.[30] SHAP is preferred over LIME for complex ensemble methods and deep neural networks because it offers a broader, mathematically rigorous perspective on feature contribution based on Shapley values.[30] SHAP assigns a value to each feature,

quantifying its contribution (positive or negative) to the model's final prediction.[30]

5.2 Translating Feature Contributions to Action

The XAI output transforms abstract model predictions into prescriptive, actionable maintenance intelligence, which is vital for achieving operational excellence.[5]

Global Interpretability: SHAP summary plots are used to identify the strategic features that, on average, drive failure across the entire dataset (e.g., the derived Stress Index, Rotational Speed, and Process Temperature). This information guides long-term strategic planning and potential modifications to operational procedures.

Local Interpretability: For every individual prediction of imminent failure, SHAP force plots reveal the specific sensor readings that contributed most strongly to that single, localized prediction.[30] This directly links the model's opaque outcome to tangible physical phenomena. For example, if a machine is predicted to fail, and the local SHAP analysis reveals that high Tool Wear and high Torque were the dominant positive contributors, the maintenance manager is informed that the machine is experiencing an Overstrain Failure.[8]

This transparent, quantifiable justification (XAI) allows the maintenance team to audit the model's recommendation and confirm that the expenditure on labor and potentially costly imported spare parts is warranted.[3] The justification serves as auditable evidence required for financial and operational accountability, ensuring the AI system fosters trust rather than skepticism.

Table 3: Translating XAI Results (SHAP) to Actionable Maintenance Decisions

High-Impact Feature (SHAP Contribution)	**Failure Type Detected**	**Technical Insight**	**Actionable Maintenance Policy for Kenyan Operations**
High/Rising Stress Index (Torque $\times$ Wear)	Overstrain Failure (OSF)	Indicates excessive mechanical load or increasing component friction, nearing the defined failure threshold.[8]	Initiate immediate load reduction or temporary speed decrease; dispatch technician for lubrication check, alignment inspection, or mechanical binding diagnosis to avert catastrophic failure.
Elevated Process Temperature (Positive Slope)	Heat Dissipation Failure (HDF)	Indicates a sustained thermal trend over the past operational window, suggesting systemic cooling	Check and clean ventilation systems or heat exchangers; verify coolant levels and flow rates; reduce machine

		failure or blockage.[17]	duty cycle until temperatures normalize.
Long Time Since Last Maintenance / High Tool Wear	Tool Wear Failure (TWF)	Component is approaching its predefined end-of-life cycle, necessitating scheduled replacement independent of transient stress events.	Schedule component replacement during the next planned downtime; immediately verify the inventory status of the high-cost imported spare part.[6]

6. System Deployment, MLOps, and Governance

Deployment requires transitioning the validated models into a scalable, reliable production environment, managed through a formal Machine Learning Operations (MLOps) framework.

6.1 MLOps Framework and Continuous Monitoring

A robust MLOps pipeline must include continuous integration and continuous deployment (CI/CD) of the model artifacts. Most critically, performance must be monitored in real time:

- **Data Drift Monitoring:** The distribution of incoming sensor data (e.g., Temperature, Speed, Torque) must be constantly tracked against the baseline data used for training. Changes in operational patterns, sensor degradation, or shifts in the extreme operating environment in Kenya [13] can cause the live data distribution to drift, making the model's predictions less accurate.
- **Model Drift Monitoring:** The predictive performance (specifically the optimized F-beta score from the Classification Pipeline) must be continuously validated on newly labeled data. If the performance degrades below a predefined operational threshold, the system must trigger an automated or manual retraining process.

6.2 Provenance Management for Accountability

In complex predictive maintenance systems, limited visibility into data processing undermines accountability and user trust.[31] To address this gap, the MLOps deployment must incorporate a rigorous **Provenance Management System**.[31]

This system should be based on an extended version of the **PROV-O schema**.[31] PROV-O provides a holistic view of data lineage, tracing not just the data transformations but also the

interactions between data, the ML model, and subsequent maintenance actions.[31]
The enhanced provenance management system must log the entire decision chain:

1. **Entity Tracking:** Recording the raw sensor data and the resulting Failure Prediction.
2. **Activity Tracking:** Documenting the Feature Engineering, Model Training, and Prediction Generation processes.
3. **Agent Tracking:** Linking the prediction and its XAI Explanation to the responsible maintenance manager (Agent) who executes the corrective action.[31]

This exhaustive audit trail ensures that every prediction is traceable, transparent, and auditable, which significantly improves user confidence, task efficiency, and system reliability, particularly when adopting new AI technologies in industrial settings.[31]

6.3 Strategic Policy Recommendations

The technical methodology enables several high-impact strategic shifts for Kenyan manufacturing:

6.3.1 Optimized Inventory and Cost Reduction

The RUL prognosis pipeline directly supports optimized inventory management. By accurately predicting the time remaining before specific, high-cost imported components require replacement, manufacturers can move away from inflexible periodic procurement schedules.[4] Instead, they can align the ordering and stocking of critical spare parts with the RUL prediction, mitigating the high capital costs and logistical delays associated with importing raw materials and spares.[6]

6.3.2 Workforce Empowerment and Trust

The integration of XAI is central to the adoption process. By making the model's inner workings transparent, maintenance staff can be trained to use the AI solution as an informed advisory tool.[3] The ability to see *why* a machine is predicted to fail allows technicians to prioritize interventions and validates their maintenance efforts, fostering institutional trust in the automated decision-support system.[32]

6.3.3 Future Scaling through Data Collaboration

While the current project uses publicly available synthetic data, securing real-world PdM datasets is notoriously challenging due to proprietary concerns and data confidentiality.[1] A strategic pathway to future scalability in Kenya involves designing the MLOps framework to be

compatible with **Federated Learning (FL)**.[33] FL is a distributed machine learning technique that allows multiple manufacturers to collaboratively train a common, robust predictive model without ever sharing their sensitive, decentralized raw sensor data.[33] This provides a mechanism for local industry to pool collective knowledge and build a high-performing model that overcomes the challenges of data scarcity and confidentiality, accelerating digital modernization efforts.[33]

7. Conclusion

The successful deployment of Artificial Intelligence for Predictive Maintenance in Kenya's manufacturing and agro-processing sectors requires a structured, multi-faceted technical approach that integrates advanced machine learning with essential engineering and operational governance. The proposed methodology, based on dual high-performance pipelines (XGBoost/LightGBM for Classification, CLSTM for RUL Prognosis) using the AI4I 2020 dataset, is designed to generate highly accurate and strategically valuable predictions. Crucially, the methodology incorporates domain-specific requirements: aggressive feature engineering based on causal discovery, rigorous handling of data imbalance by optimizing the F-beta score for Recall, and proactive sensor anomaly detection to address harsh environmental factors. The entire system is anchored by Explainable AI (SHAP) and a PROV-O based Provenance Management System, ensuring that predictions are not only accurate but also auditable, trustworthy, and actionable.
By moving beyond simple predictive models to an integrated, explainable decision-support framework, Kenyan manufacturers can strategically leverage AI to reduce unplanned downtime, maximize equipment longevity, and achieve significant cost savings in spare parts and operational efficiency, mirroring the successful digital transformation observed in other critical sectors, such as grid management and finance.[32]

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